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Analysis of water-permeable fractured zone in weakly cemented overburden considering rock strain-softening

Sen Xue[1,2], Qiqing Wang[2] & Zhen Song[1,3]✉

The accurate prediction of the water-permeable fractured zone (WFZ) height is critical for preventing water inrush disasters in longwall mining, especially in ecologically fragile areas. Current methods for evaluating the WFZ often neglect the strain-softening behavior of rock, leading to a commonly observed deviation in predictions. This study investigates the overburden failure mechanisms and WFZ development by integrating in-situ monitoring, empirical analysis, and numerical modeling using the Mohr-Coulomb (MC) model and an advanced Strain - Softening (SS) model. The SS model accurately replicated the progressive failure of the overburden, predicting a WFZ height of 144.5 m, which closely aligned with the in-situ data. In contrast, the MC model, with its assumption of constant post-peak strength, underestimated the height by 16 m, yielding a result of 128.5 m. This discrepancy concludes that the strain-softening behavior of rock is a dominant factor controlling WFZ development. Relying solely on traditional empirical formulas or simplistic constitutive models may lead to non-conservative underestimations of the WFZ, thereby escalating mining safety and ecological risks. The validated SS model provides a superior tool for realistic water hazard assessment and strategic mine planning.

Keywords Coal mining, Water-permeable fractured zone, Mohr-coulomb model, Strain softening

Coal mining activities in China's western arid and semi-arid ecologically fragile regions are commonly conducted in the shallow-buried seams with thin bedrock and thick unconsolidated overburden[1]. Longwall extraction can induce progressive deformation and failure of the overlying strata, forming a well-developed water-permeable fractured zone (WFZ) that may hydraulically connect shallow water-bearing units (e.g., loose sand layers and weakly cemented sandstone aquifers) with underground workings[2,30]. Such connectivity often substantially disturbs groundwater recharge-flow-discharge regimes, causing regional groundwater-level decline and increasing the likelihood of mine water inflow; in extreme cases, water-sediment inrushes may occur and threaten operational safety[3–5]. Meanwhile, mining-induced fracturing may exacerbate environmental problems such as soil erosion, vegetation degradation, and groundwater contamination[6–8]. Therefore, accurate prediction of WFZ height is a prerequisite for water-hazard assessment and environmentally responsible mine planning under confined aquifer conditions.

Characterizing the WFZ evolution has long been a focus in mining hydrogeology. In China, the empirical formula issued by the former State Bureau of the Coal Industry is still widely used to estimate WFZ height. However, its roof-lithology classification, i.e., hard, medium-hard, soft, extremely soft, is overly coarse and cannot adequately reflect the heterogeneity of overburden strata[7,9]. Moreover, this formula was calibrated mainly with field data from Carboniferous-Permian coalfields in eastern China, and its transferability to western mining areas with weakly cemented Jurassic strata and strong mining disturbance remains uncertain. Consequently, various approaches were developed, including theoretical analysis[10,11], physical similarity simulation[12], in-situ/advanced monitoring[13], data-driven methods[14], and numerical modeling[15]. Among them, numerical simulation is particularly attractive because it can reproduce the spatiotemporal evolution of overburden fracturing and enables sensitivity analyses of key controlling factors under different mining scenarios. Nevertheless, the predictive reliability of numerical simulations is strongly controlled by how accurately the stress-strain behavior of rocks is represented, i.e., the selection of an appropriate constitutive model.

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In western Chinese mining areas, coal measures are younger and poorly lithified, with pronounced weak cementation, often dominated by argillaceous components[1,16,17]. Previous investigations[18–24] show that such weakly cemented rocks exhibit low strength and distinct post-peak responses, including strain-softening and dilatancy, and their mechanical behavior is highly sensitive to confining stress and water saturation. For example, Lv et al.[25] found that pore-pressure and water effects can induce pronounced post-peak degradation that is well captured by strain-softening laws (e.g., exponential decay of strength parameters with plastic shear strain), and the associated plastic or softening zones can be independently identified using micro-CT observations. Moreover, both static and dynamic tests consistently show that increasing water content significantly reduces rock strength and stiffness and alters the post-peak stress-strain response, indicating a moisture-dependent damage and softening process across different loading regimes[26,27]. These features imply progressive damage accumulation and evolving stiffness/strength during deformation, which can fundamentally govern fracture propagation in the overburden and hence the WFZ development. Although strain-softening behavior is crucial for WFZ development, the corresponding research has largely remained at the laboratory or unit scale, and its integration into engineering-scale simulations is still limited. By contrast, WFZ evolution involves pronounced large deformation, progressive fracture propagation, and strong rock-mass discontinuity, for which discrete element methods are well suited. However, most discrete element studies of WFZ development still adopt conventional Mohr-Coulomb or strain-hardening constitutive formulations, with little explicit consideration of strain-softening. This methodological mismatch restricts accurate reproduction of the full stress-strain response of weakly cemented strata and may therefore bias WFZ height predictions.

In this study, a longwall panel in the Shendong mining area is investigated to clarify the role of strain-softening behavior in the WFZ development. In-situ monitoring is combined with numerical simulation, in which a strain-softening constitutive model is implemented and systematically compared with the conventional Mohr-Coulomb model. This integrated field-numerical framework enables quantitative evaluation of WFZ evolution and height prediction under weakly cemented overburden conditions.

Engineering background

Mining condition

Bu Lianta Coal Mine is one of the typical coal mines in the Shendong mining area, located in Ejin Horo Banner, Ordos City, Inner Mongolia Autonomous Region (Fig. 1). The mine spans an area of 106.43 km², and the main coal seams include 1–2, 2–2, and 3 – 1 coal seams, with a recoverable reserve of 1.345 billion tons. The 31,401

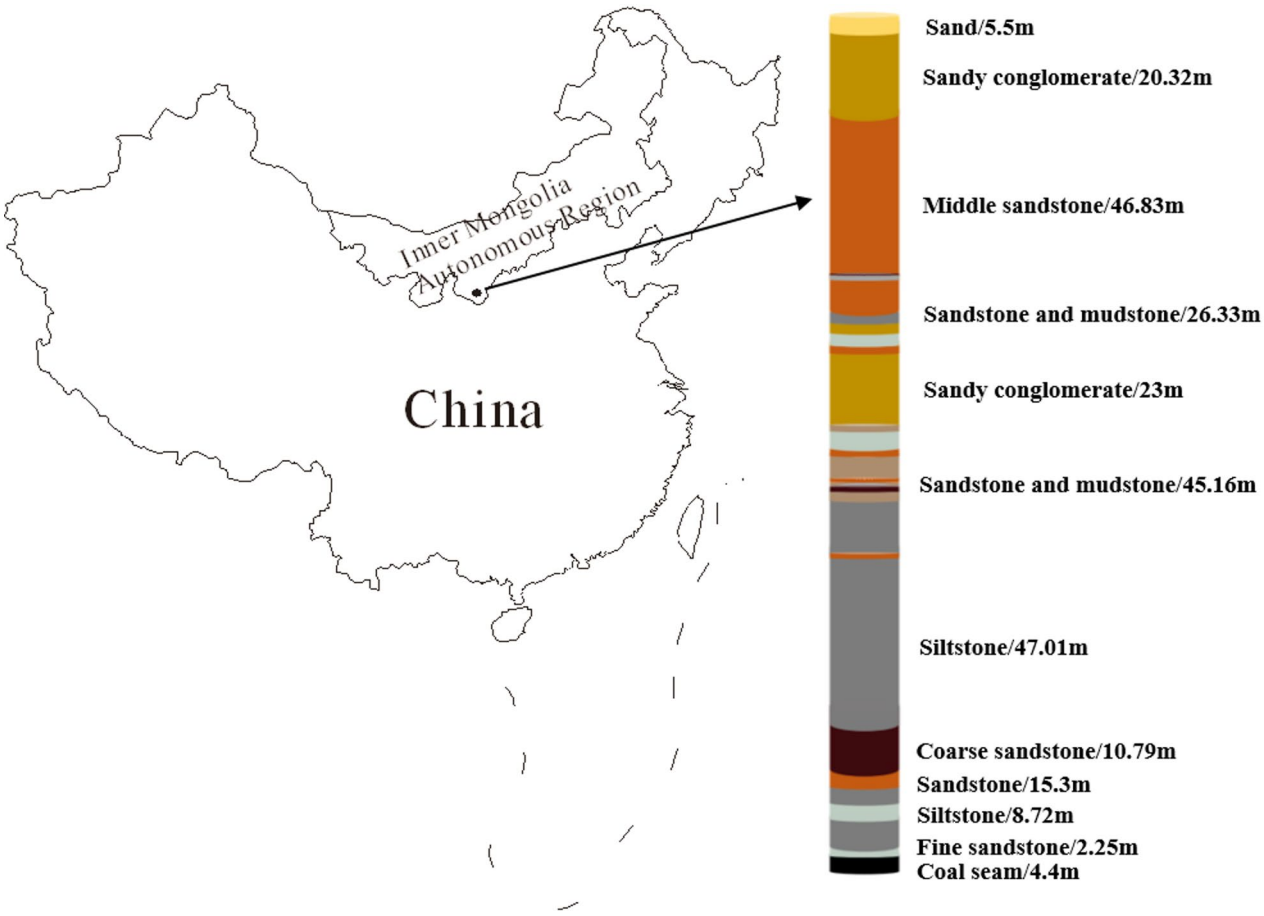

Fig. 1. Study area location map and geological profile.

working face is the first mining face in the four panel area, with a mining slope length of 265 m and a strike length of 4629 m. The coal seam being mined is the 1–2 seam, buried at depths ranging from 180 m to 250 m, with a mining thickness between 4.2 m and 4.5 m, and a coal seam inclination angle of 1° to 3°. It is classified as a nearly horizontal coal seam and employs a comprehensive mechanical mining technique known as the full-height strike longwall retreat full-collapse method for single thick seam extraction. The working face is predominantly covered by Quaternary loose aeolian sand, with thicknesses varying from 5 m to 25 m. The main roof is predominantly composed of coarse and medium sandstone, typically exceeding 20 m in thickness, relatively hard, and cemented with mud and calcium. The immediate roof consists of sandy mudstone, siltstone, or fine to medium sandstone, ranging in thickness from 1.5 m to 7.16 m, often forming thick layers, cemented with mud, and semi-rigid.

Existing empirical formulas for WFZ calculation

Through regression analysis of the measured results of multiple coalfields in North China, Liu[28] proposed an empirical formula for the height of WFZ. Those formulas have been widely adopted in China was subsequently incorporated into the "Coal mining regulations under the buildings, water and rails". Coal seam roof structures are categorized into four types (hard, medium-hard, weak, and extremely weak) according to engineering practice and the uniaxial compressive strength of the rock. It was determined through a comprehensive assessment that the roof at 31,401 working face within the "medium-hard" type, based on the site's engineering geological conditions. The mining thickness at 31,401 working face was 4.4 m, and the height of WFZ can be calculated as follows according to the empirical formula of Liu[28]:

$$h_w = \frac{100M}{1.6M + 3.6} \pm 5.6 = 41.4 \pm 5.6m$$

where h_w is the WFZ height measured from the coal seam roof, M is the mining height, and 100, 1.6, and 3.6 are empirical constants derived from field observations.

In-situ monitoring for WFZ

The most straightforward and precise approach to whether fractures in the overburden are water-permeable is to inject water into them. In this study, the height of WFZ is measured using flushing fluid leakage observations and borehole water level observations. As shown in Fig. 2, one borehole was drilled within unmined area, and one borehole was drilled in the goaf.

During the in-situ investigation of the development height of WFZ, Z1 borehole is situated in the unmined area of the working face. The leakage rate of the drilling flushing fluid and the water level within the borehole directly indicate the development degree of primary fractures in the overlying strata of the roof. Figure 3 shows the detection results of Z1 borehole. During the drilling process, the leakage rate curve of the flushing fluid in Z1 borehole was mostly between 0.0035 ~ 0.0075 L/s·m, with no circulation interruptions observed. It only slightly increased in some areas, and the starting points of the increasing trend sections were 92.68 m, 104.37 m, and 121.98 m. In the meantime, the burial depth of the drilling water level ranges from 0.73 to 6.58 m. The results indicating that the primary fractures inside the bedrock layers in the 31,401 working face were not abundant.

The change of washing fluid leakage quantity and water level of drilling Z2 is shown in Fig. 4. Before drilling to a depth of 96.96 m, there was no significant fluctuation in the leakage curve of the flushing fluid; During the stage of drilling depth from 96.96 m to 97.46 m, the consumption of drilling flushing fluid rapidly increased from 0.0065 L/s·m to 29.9629 L/s·m. there has been a 3900-fold increase compared to pre-mining levels. The

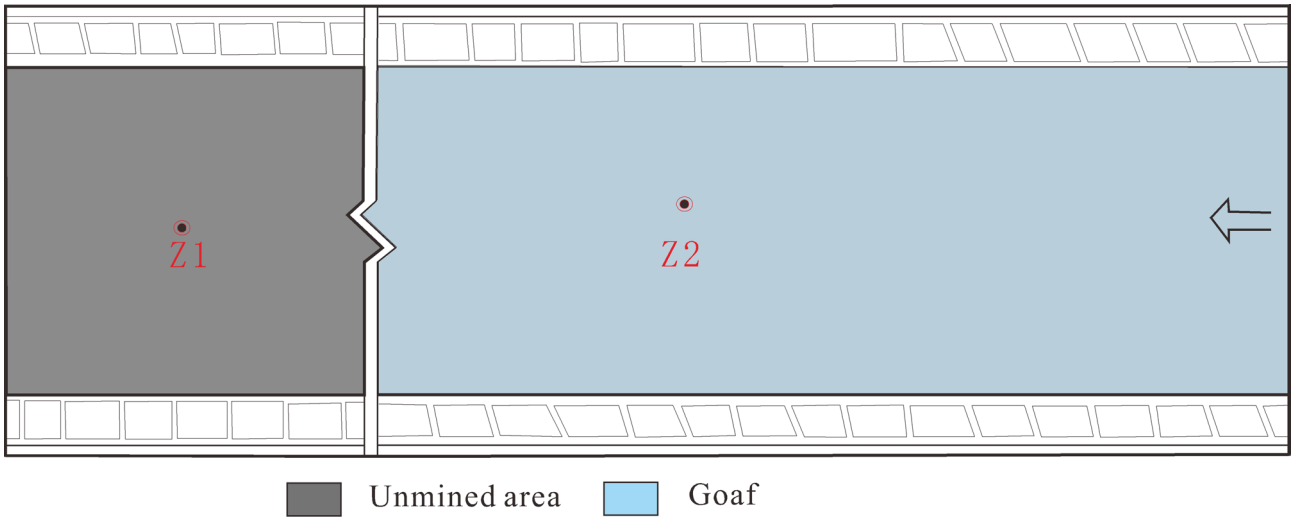

Fig. 2. Borehole location of 31,401 working face.

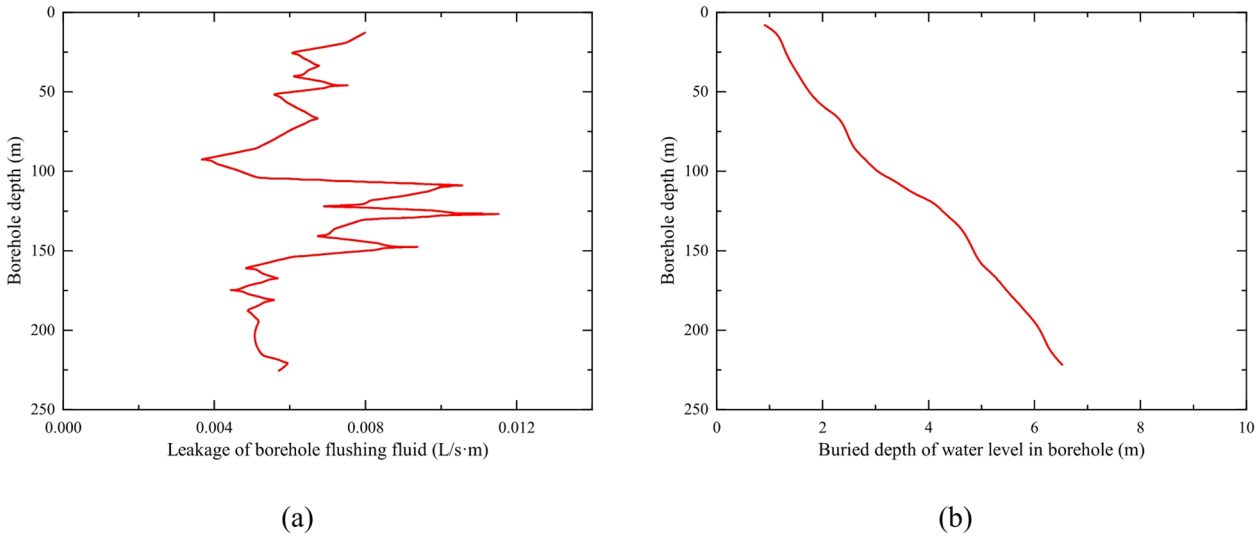

Fig. 3. Curves of washing fluid leakage quantity and water level of drilling Z1. (**a**) washing fluid leakage quantity, (**b**) water level in borehole.

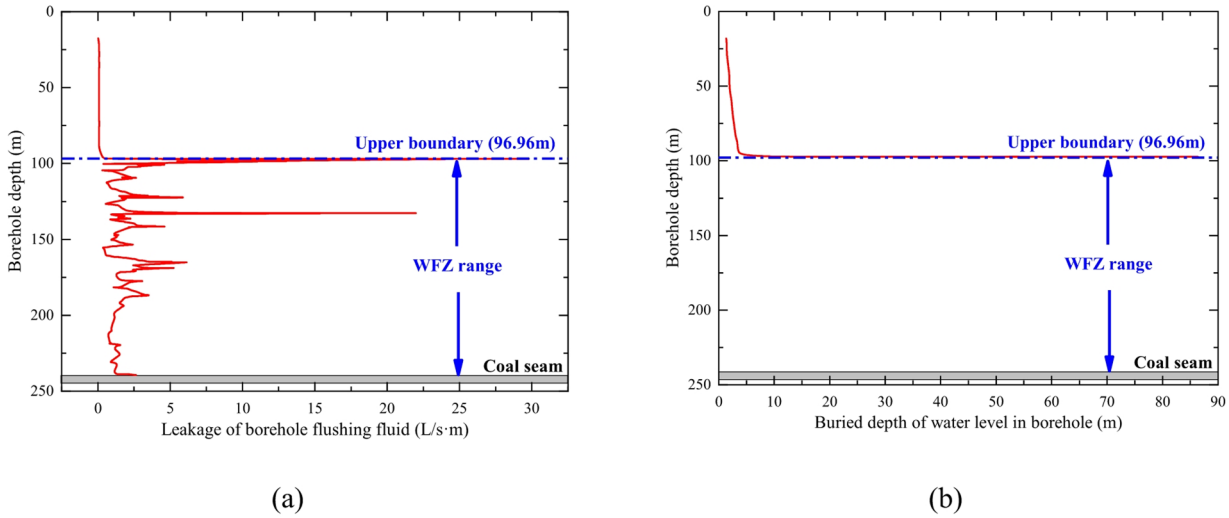

Fig. 4. Curves of washing fluid leakage quantity and water level of drilling Z2. (**a**) washing fluid leakage quantity, (**b**) water level in borehole.

water level in the Z2 hole instantly leaked from 3.85 m to 97.46 m at the bottom of the hole, at which point all the drilling flushing fluid was lost. The onset of a distinct upward trend in fluid loss rate serves as a key diagnostic criterion for delineating the upper interface of the fractured zone. This hydromechanical response provides direct evidence of the intensive fracturing and permeability enhancement within the zone. In this study, the top interface of the WFZ is the depth of 96.96 m, and the height of that is 142.6 m.

Figure 5 shows the displays the borehole log imagery from different depths by borehole video observation. At shallow depths (above 65 m), it is in a pristine condition, devoid of discernible cracks, which points to a region safely above the primary influence of the mining excavation. The first evidence of damage emerges at 76 m, where characteristic ring-like fractures appear, acting as precursors to more widespread failure. As depth increases further, this initial damage cascades into a complex and pervasive fracture system. Characterized by high-density, extensively interconnected, and crisscrossing networks, these fractures indicate a severe failure state that stands in stark contrast to the mildly disturbed overlying strata, underscoring the intense mechanical disruption experienced at these levels. The core from borehole Z2 shows a progressive degradation of the rock mass with depth.

The imaging results further illustrate how mining operations fundamentally alter the native groundwater system. Pre-mining data from borehole Z1, reflected in its washing fluid leakage curve, show that the roof strata possess sparse primary fractures but serve as a weakly cemented framework that is vulnerable to stress. The

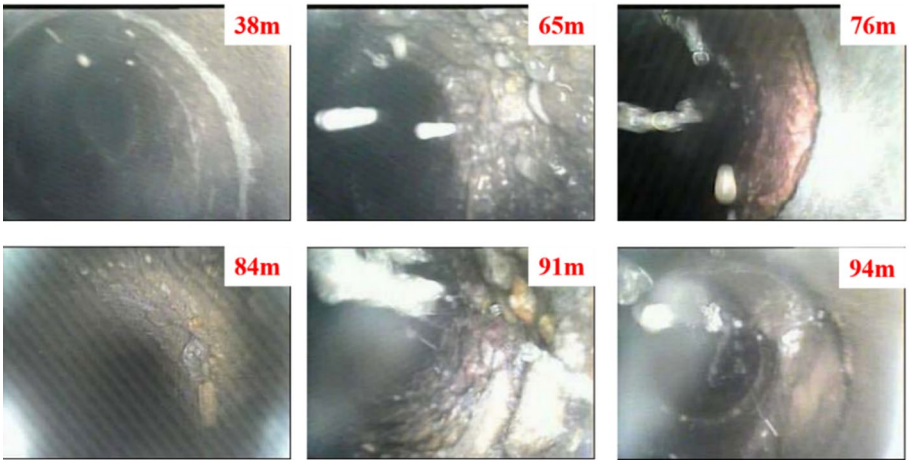

Fig. 5. Borehole video observation results at different depth of drilling Z2.

introduction of mining-induced stresses triggers structural failure in these strata. This process, governed by their poor cementation, leads to the development of secondary porosity and fracture networks, which in turn significantly increase the hydraulic conductivity and permeability of the rock mass.

In general, mining-induced rock strength degradation generates dense, interconnected fractures from initially intact strata. This evolution of secondary porosity significantly enhances hydraulic conductivity, fundamentally altering the groundwater regime.

Discrete element analysis

Strain softening model of rock

The Mohr-Coulomb (MC) model is a fundamental and widely adopted elastoplastic constitutive model comprising a linear elastic component and an ideal plastic component. The material response is linearly elastic until the stress state satisfies the MC failure criterion, which defines the onset of yielding and subsequent plastic flow. This yield surface of the MC model is universally defined by three stress invariants, with its mathematical representation provided below[29]:

$$f = R_{mc}q - p\tan\varphi = 0 \tag{1}$$

$$R_{mc} = \frac{1}{\sqrt{3}\cos\varphi}\cos\theta + \frac{1}{3}\sin\theta\tan\varphi \tag{2}$$

where p, q is the mean effective normal stress and generalized shear stress, respectively. φ is the friction angle. R_{mc} is the deviatoric stress measure defined as a function of friction angle φ and the Lode angle θ. The yield surface of MC model in the three-dimensional principal stress space is shown in Fig. 6(a).

As a unique geological material, rock contains a certain amount of randomly distributed micro-pores and cracks of varying shapes, resulting from long-term and complex diagenetic processes. This internal structure confers characteristics such as heterogeneity and anisotropy. In fact, due to the influence of internal structure, after the stress reaches the peak strength, the strength of rock rapidly decreases to a low level with the continuous increase of deformation. This phenomenon of rock material performance degradation caused by deformation is called strain softening.

The modification of the traditional MC model is therefore necessary to enhance its applicability. The schematic of this modification process is illustrated in Fig. 6b, i.e., the strain softening (SS) model. The complete mechanical response of the material is defined by the continuous translation and contraction of its yield surface from an initial (Point A) to a final (Point B) state in the stress space. After the peak stress is attained, the rock material exhibits strain-softening, a direct consequence of internal structure degradation such as the progressive development of micro-cracks or bond breakage. This is macroscopically observed as a stress decrease with ongoing plastic deformation. Concurrently, the yield surface contracts inward in stress space, reflecting the material's diminishing strength and hardening capacity. Failure is mechanistically defined as the point where the yield surface ceases to contract, stabilizing into its final form, termed the residual failure surface. The attainment of this surface signifies the initiation of a perfectly plastic flow state, where deformation continues without any further change in stress, indicating that the rock material's internal shear resistance has been fully exhausted. This residual failure surface plays a critical role in predicting the height of WFZ in overlying strata.

Specifically, in the numerical simulation, once an element yields, its new shear strength parameters are computed at each iteration based on the magnitude of the plastic strain. This update is performed using a nonlinear function that relates the shear strength parameters to the plastic strain. The model then proceeds to the next iteration with the updated parameters. This iterative mechanism allows for the representation of the strain-softening behavior of rock. The strength parameter softening process of SS model are shown in Fig. 7, which can be expressed as follows:

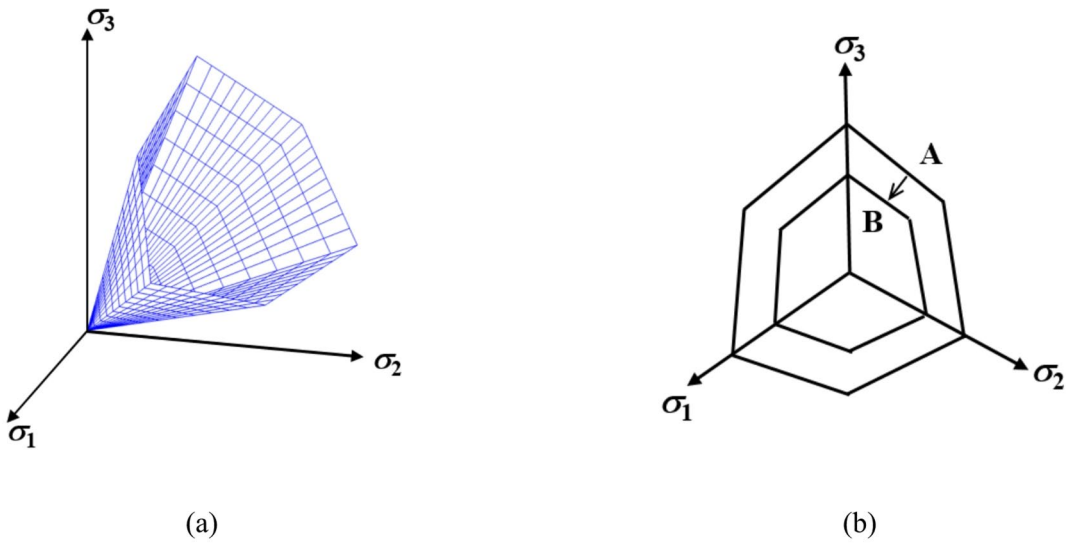

Fig. 6. Yield surface of MC model and SS model. (**a**) MC model in principal stress space, (**b**) SS model in π plane.

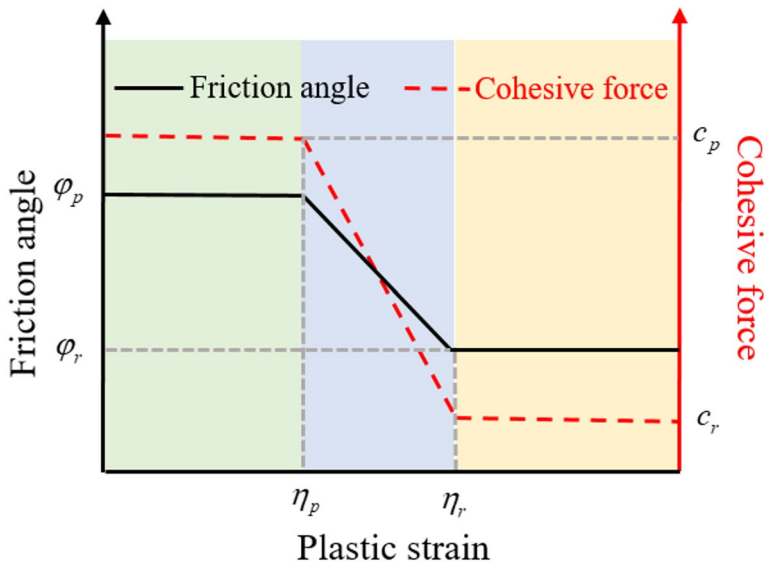

Fig. 7. Softening process of strength parameter.

$$c = \begin{cases} c_p \ , & \eta \leqslant \eta_p \\ c_p - \frac{c_r - c_p}{\eta_p - \eta_r}(\eta - \eta_p) & , \quad \eta_p < \eta \leqslant \eta_r \\ c_r \ , & \eta > \eta_r \end{cases} \tag{3}$$

$$\tan\varphi = \begin{cases} \tan\varphi_p \ , & \eta \leqslant \eta_p \\ \tan\varphi_p - \frac{\tan\varphi_r - \tan\varphi_p}{\eta_p - \eta_r}(\eta - \eta_p) & , \quad \eta_p < \eta \leqslant \eta_r \\ \tan\varphi_r \ , & \eta > \eta_r \end{cases} \tag{4}$$

where c_p, c_r is the peak and residual cohesive force, respectively. φ_p, φ_r is the peak and residual friction angle, respectively. η_p, η_r is peak and residual plastic strain, respectively.

Numerical model

In mining engineering, numerical modeling represents the most common approach for investigating the stability and failure patterns of overburden strata. The field primarily utilizes two computational frameworks:

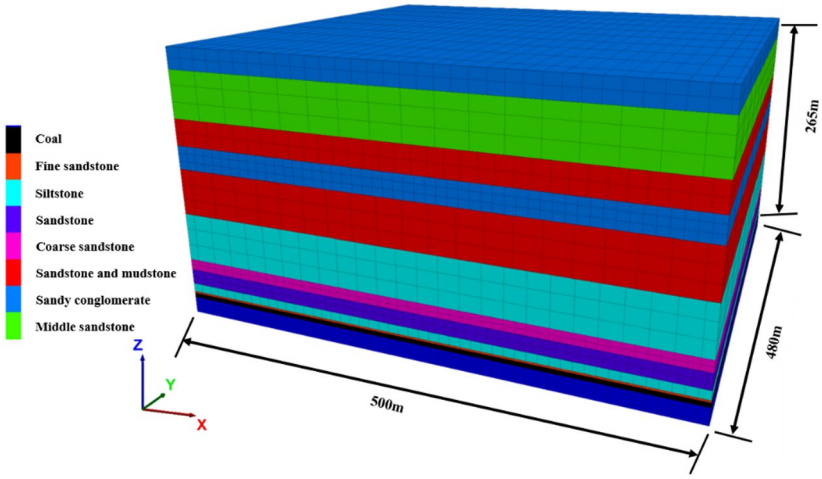

Fig. 8. Numerical model of the 31,401 working face.

Rock strata	Bulk modulus (GPa)	Shear modulus (GPa)	Peak Cohesion (MPa)	Residual Cohesion (MPa)	Peak Friction angle (°)	Residual Friction angle (°)	Tensile strength (MPa)
Coal	3.5	1.7	2.65	1.5	30	16	2.39
Siltstone	28.5	22.8	4.18	2.1	50	27	5.41
Fine sandstone	49.3	38.1	5.57	2.7	54	30	6.45
Sandstone	33.4	27.1	4.98	2.4	46	25	6.10
Coarse sandstone	40.6	33.2	6.13	3.3	50	27	5.24
Sandstone and mudstone	22.5	18.2	3.57	1.9	35	28	2.57
Sandy conglomerate	30.4	26.5	5.87	2.8	39	22	4.54
Middle sandstone	34.1	27.6	7.3	3.5	48	30	5.2

Table 1. Physico-mechanical properties of coal, rocks.

methods rooted in continuum mechanics (e.g., FEM, FDM) and those based on discontinuum mechanics (e.g., DEM). Although continuum methods are widely applied for their efficiency in simulating elastic and plastic deformation, they face significant limitations in capturing the post-failure response of rock masses. Mining-induced strata movement is characterized by a complex geomechanically transition from initial, continuous deformation to the progressive development of discontinuous fractures and structural collapse[7,8]. This process, governed by the initiation, propagation, and interconnection of fractures, falls outside the theoretical scope of conventional continuum models. To overcome this limitation, the three-dimensional distinct element code (3DEC) was employed. As an advanced DEM-based approach, 3DEC is uniquely capable of modeling the collective behavior of a fractured rock mass as an assemblage of discrete, interacting blocks. This makes it a proven and appropriate tool for simulating the key mechanisms of overlying strata failure, including caving, bending, and separation, in response to underground mining activities. In this section, to clarify the influence of rock strain-softening on the WFZ height of 31,401 working face, the development of WFZ is analyzed by SS model and MC model.

Figure 8 shows the numerical model of the 31,401 working face. The overlying strata was divided into rock stratum consistent with the strata's thickness. The rock stratum was discretized into blocks by setting the joints in the strata. The length, width, and height of the 3DEC model were 500 m, 480 m, and 265 m, respectively. The mining thickness, mining length, and mining width were 4.4 m, 300 m, and 265 m, respectively. The excavation was performed in 5 m intervals. A vertical load was applied to the top of the bedrock to simulate the weight of the 5.5 m unconsolidated layers. The vertical displacement at the base and horizontal displacement along the sides are limited, and the upper boundary is a free boundary. Based on laboratory results and empirical approaches, the mechanical parameters of the rock and coal used in this numerical simulation are listed in Tables 1 and 2.

Numerical simulation results

Distribution characteristics of the plastic zone

The distribution characteristics of the plastic zone in the surrounding rock serve as a critical indicator for predicting the failure mode and extent of the overlying strata. Figures 9, 10 and 11 depict the numerical model cross-sections under different extraction distances. Based on a detailed analysis of the observation cross-section, the evolutionary process of the overburden failure can be clearly identified. Under the combined effects of

Rock strata	Normal stiffness (Gpa)	Shear stiffness (Gpa)	Cohesion (Mpa)	Friction angle (°)	Tensile strength (Mpa)
Coal	1.2	1.4	0.13	10	2.3
Siltstone	40	33	0.71	23	3.2
Fine sandstone	45	36	0.96	27	1.5
Sandstone	33	24	1.05	29	2.0
Coarse sandstone	42	31	1.36	33	1.8
Sandstone and mudstone	19	12	0.85	21	4.5
Sandy conglomerate	34	28	0.74	25	2.6
Middle sandstone	35	24	0.63	18	3.5

Table 2. Physico-mechanical properties of coal and rocks joints.

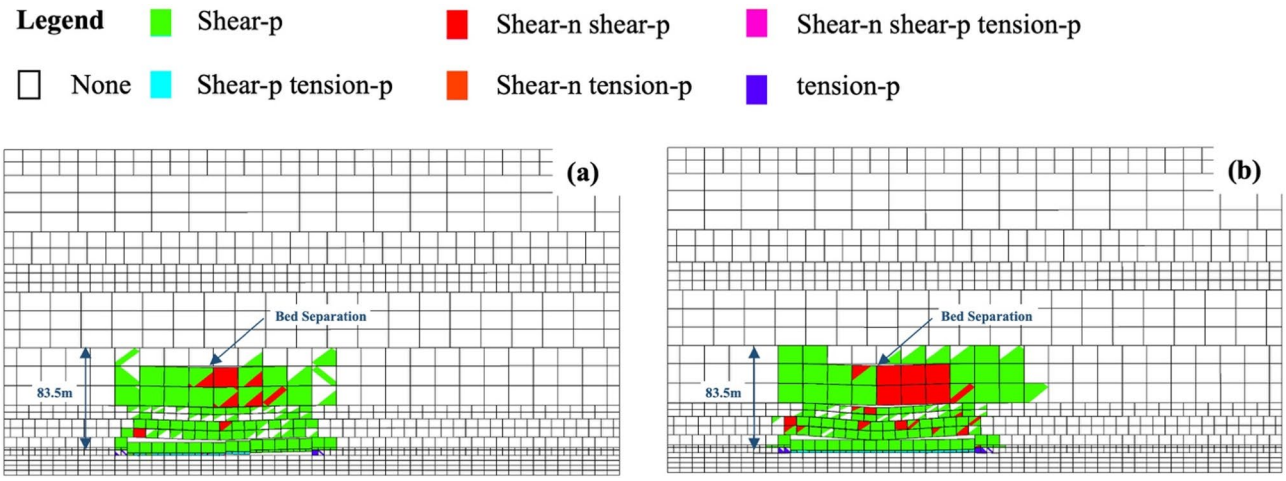

Fig. 9. The plastic zone of the MC model and SS model (mining distance of 150 m). (**a**) MC model and (**b**) SS model.

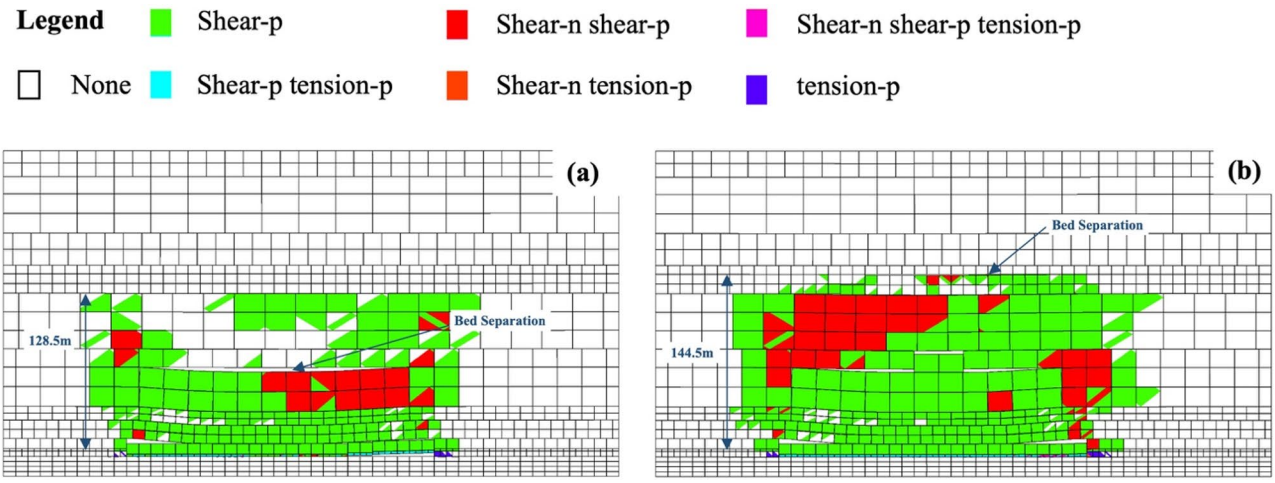

Fig. 10. The plastic zone of the MC model and SS model (mining distance of 250 m): (**a**) MC model and (**b**) SS model.

self-weight stress and mining-induced redistribution, the immediate roof of the goaf initially experienced bed separation and bending deformation. Subsequently, tensile and shear failures progressively developed in the overlying strata, leading to the formation of a plastic zone that was predominantly concentrated at the roof arch and arch foot areas, exhibiting a broadly symmetrical distribution pattern. Notably, the development of the plastic zone was significantly influenced by pre-existing joints and fractures, with noticeable intensification

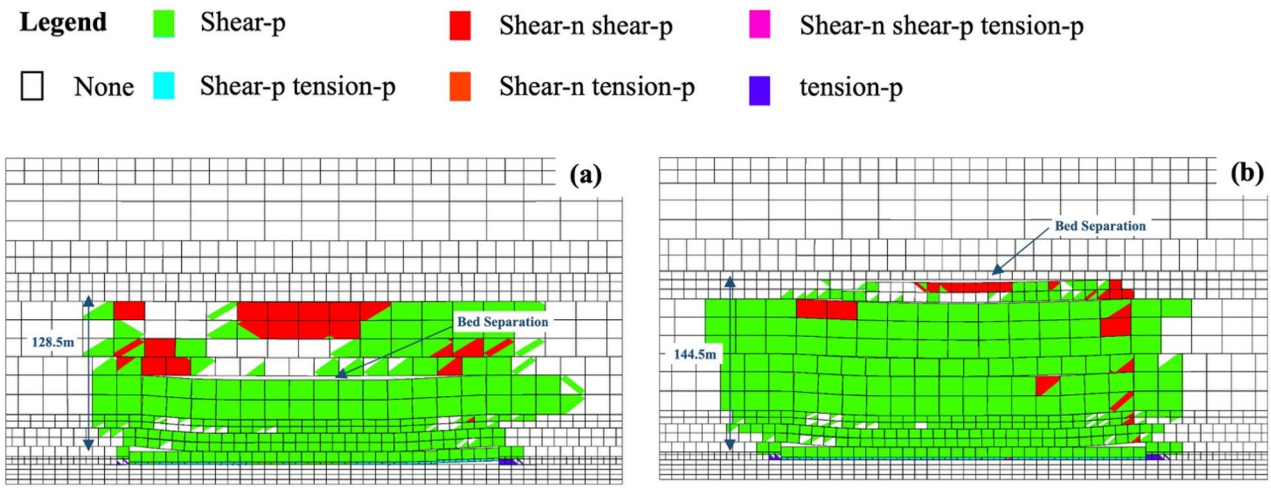

Fig. 11. The plastic zone of the MC model and SS model (mining distance of 300 m): (**a**) MC model and (**b**) SS model.

observed in the vicinity of these structural discontinuities. This localization of damage underscores the critical role of structural weaknesses in controlling the failure mechanisms of the surrounding rock mass.

In addition, analysis of the plastic zone distribution reveals that the roof strata are predominantly characterized by shear and tensile plastic zones, indicating a composite failure mechanism involving both shear and tensile damage modes. By integrating the evolutionary trends of plastic zones observed in Figs. 9, 10 and 11, it can be reasonably inferred that coal mining activities inevitably induce irreversible perturbations in the surrounding rock of the roof. These perturbations not only alter the displacement and stress distribution within the roof but also drive progressive expansion of the plastic zones, significantly increasing the risk of roof collapse.

Effect of strain softening

Figures 9, 10 and 11 also demonstrate the differences between the MC model and the SS model simulation results, which are evident. Figure 9 indicates that at a working face advance of 150 m, the simulated WFZ heights from both the MC and SS models are consistent at 83.5 m; nevertheless, the SS model predicts a broader spatial distribution of the WFZ and more evident bed separation.

At a critical working face advance of 250 m, as shown in Fig. 10, a stage characterized by the rapid expansion of the overlying plastic zone, the simulation outcomes of the MC and SS models diverge fundamentally due to their distinct constitutive representations of post-failure behavior. The conventional Mohr-Coulomb (MC) model, which does not account for rock strain softening, depicts a failure mode dominated by lower-strata caving with minimal upper strata disruption. In this idealized elastoplastic framework, the rock strength remains constant after yielding, thus preserving the integrity of the upper strata and allowing pre-existing bed separations to propagate laterally. This results in a water-conducting fracture zone (WFZ) developing to a height of 128.5 m. In stark contrast, the Strain-Softening (SS) model incorporates the critical mechanism of rock strength degradation after the peak strength is reached. This constitutive formulation more accurately captures the progressive failure and strength loss in the overburden. It predicts pronounced subsidence and extensive failure penetrating the upper strata, as the weakened rock mass undergoes continued deformation. This widespread strength degradation induces a compaction effect that closes prior separations, while simultaneously triggering the generation of a new, high-position separation at a greater depth due to differential deformation across weakened interfaces. Consequently, this more realistic simulation of progressive failure culminates in a significantly greater WFZ height of 144.5 m. The 16-meter discrepancy in the predicted WFZ height is therefore not a numerical anomaly but a direct manifestation of whether strain-softening behavior is considered. Following further advance of the working face, the maximum development height of the WFZ ceased to increase upon the overburden reaching full excavation.

The WFZ height changes by the MC and SS model are shown in Fig. 12. The WFZ height predicted by both models appears to more accurate, comparison with the prediction zone of empirical formula. SS model has the highest prediction accuracy (error is about 1.3%), in addition, MC model underestimate the WFZ height when compared to this measured results. Throughout the evolution of the WFZ, the MC model yielded consistently lower predictions than the SS model. This discrepancy underscores that the MC model's assumption of a constant post-peak strength results in an underestimation of both fracture propagation and vertical extent. In contrast, the SS model, which accounts for strength degradation, predicts a wider and more complex failure zone, thereby providing critical insight for accurate assessment of water inrush risks from overlying aquifers.

Conclusion

This study conducted a comprehensive evaluation of overburden failure and the development of the WFZ by integrating in-situ monitoring, empirical analysis, and numerical modeling with distinct constitutive laws. The main conclusions are summarized as follows: (1) In-situ investigations based on borehole fluid leakage tests

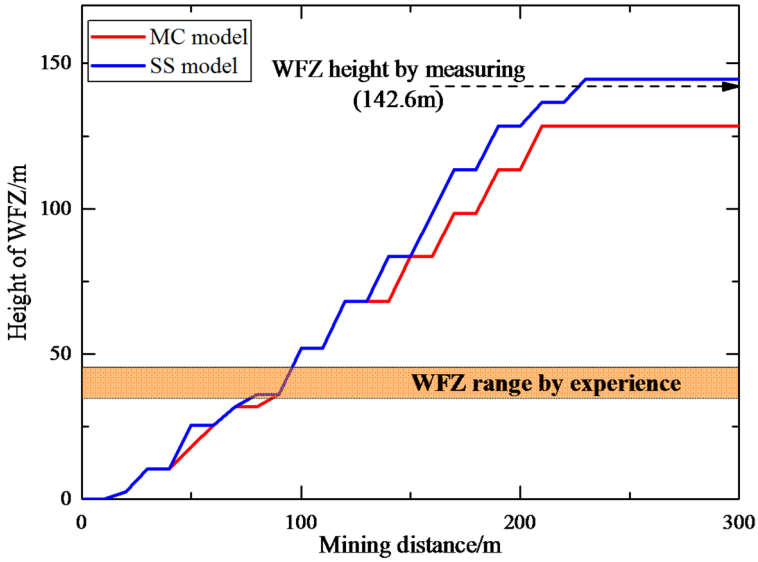

Fig. 12. WFZ height changes by the MC and SS model.

and borehole video observations clearly identified the development height of WFZ, which was determined to be 142.6 m. (2) The conventional empirical formula significantly underestimated the WFZ height under weakly cemented overburden conditions, potentially leading to non-conservative assessments and elevated risks to mining safety and the local ecological environment. (3) Numerical simulations performed with 3DEC successfully reproduced the progressive failure behavior of the overburden strata. The strain-softening constitutive model showed good agreement with the in-situ observations, whereas the conventional Mohr-Coulomb model consistently underestimated the final WFZ height. (4) The results confirm that the choice of constitutive model plays a critical role in WFZ prediction. By explicitly accounting for post-peak strength degradation, the strain-softening model provided a more realistic representation of overburden failure and yielded a WFZ height prediction of 144.5 m, closely matching the field measurements.

Future work will focus on integrating the micro-mechanical origins of strain-softening behavior, through detailed laboratory investigations of weakly cemented rocks (e.g., scanning electron microscopy observations and mineralogical analyses), to further constrain constitutive model parameters and strengthen the physical basis of the numerical simulations.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions
Sen Xue: Writing - original draft; review and editing; Software and coding; Funding acquisition. Qiqing Wang: Writing - review and editing; Data curation; Validation. Zhen Song: Funding acquisition; Supervision.

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Declarations

Competing interests
The authors declare no competing interests.

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